# E-commerce & Finance Insights Report

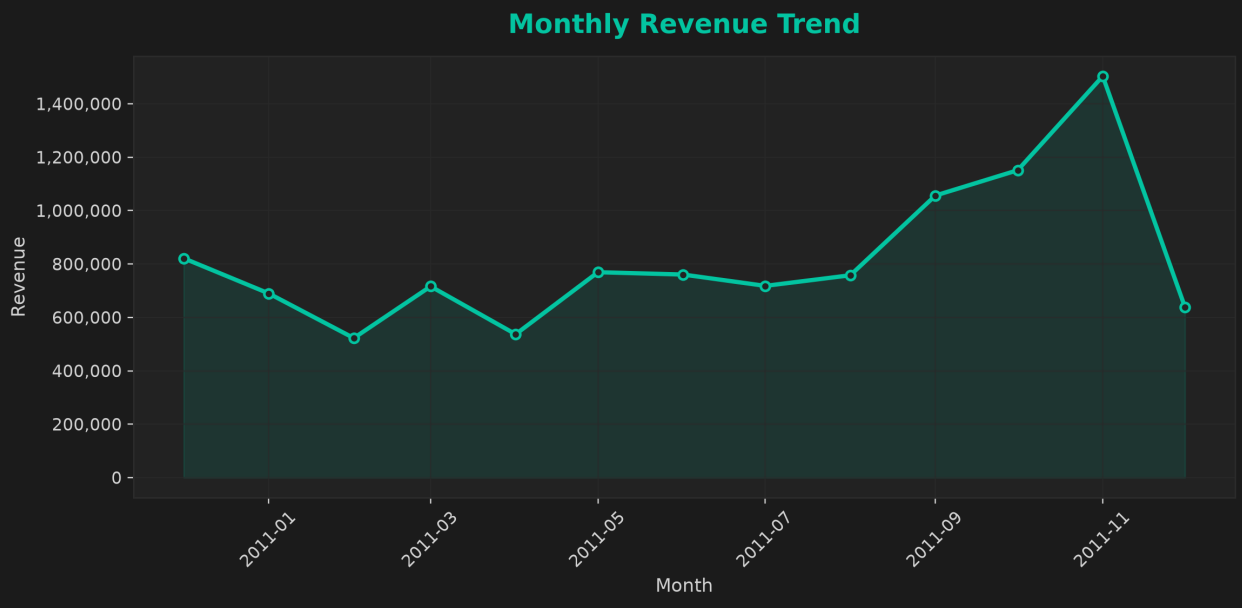
Generated via Python ReportLab • Author: Huzeif Khan

## Key Performance Indicators (KPIs)

- **Total Revenue**: 10,641,558.95
- **Total Customers**: 4,338
- **Average Order Value**: 2,453.10

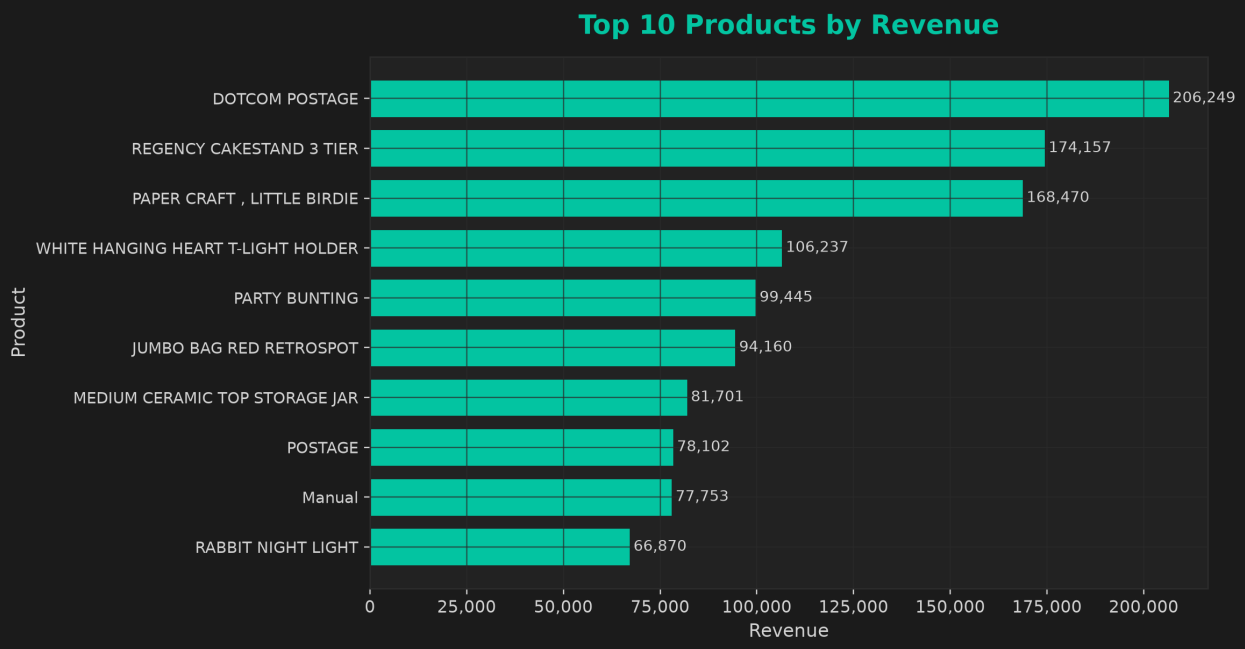
## Visual Summary

### Monthly Revenue Trend

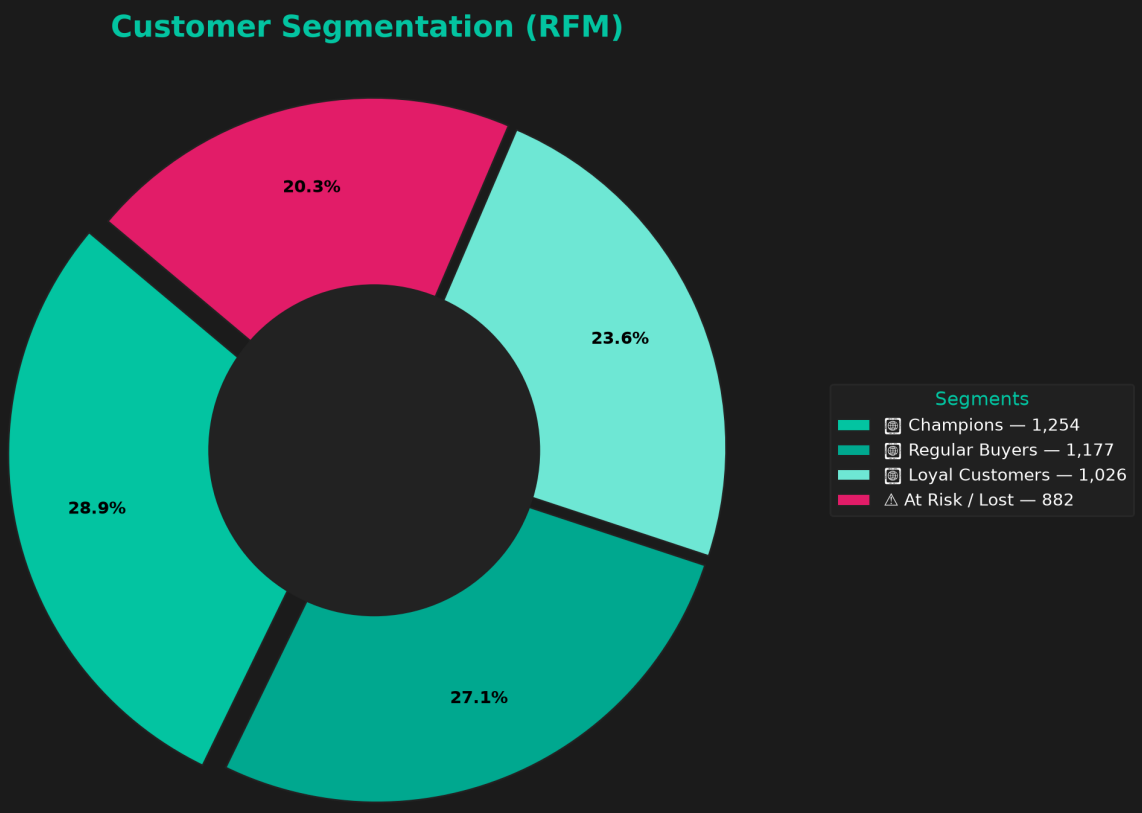


# Top 10 Products by Revenue

## Top 10 Products by Revenue

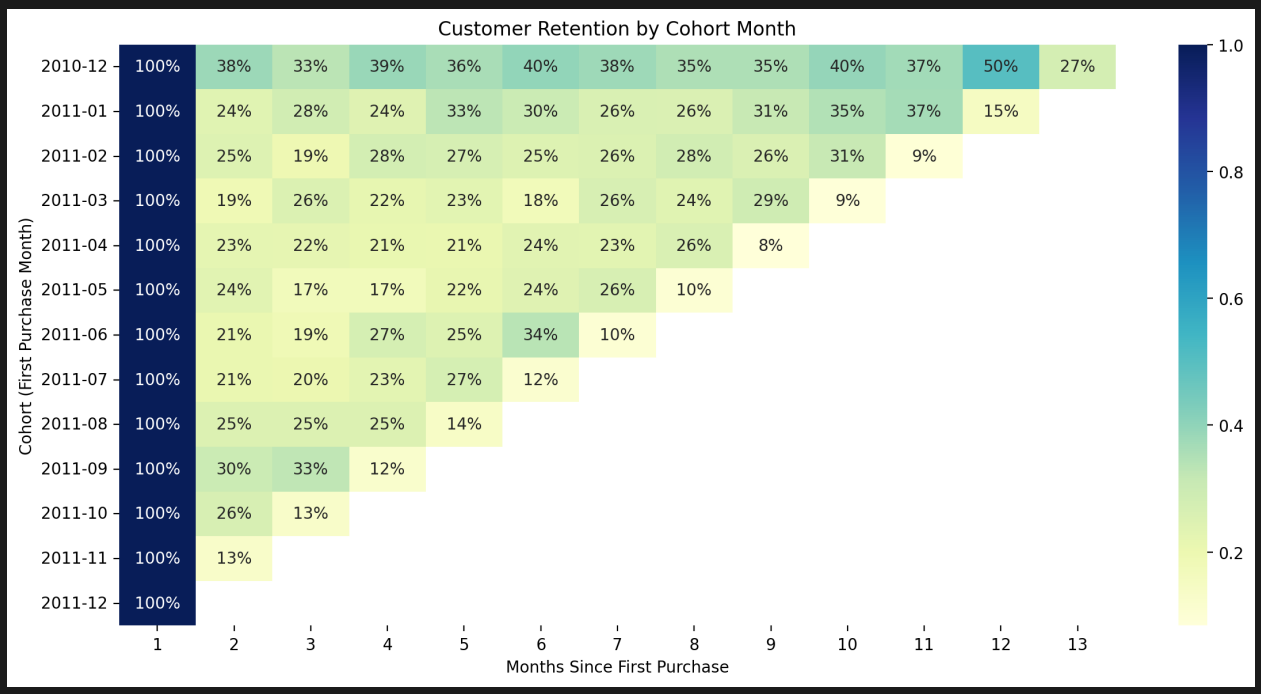


# Customer Segmentation (RFM Model)



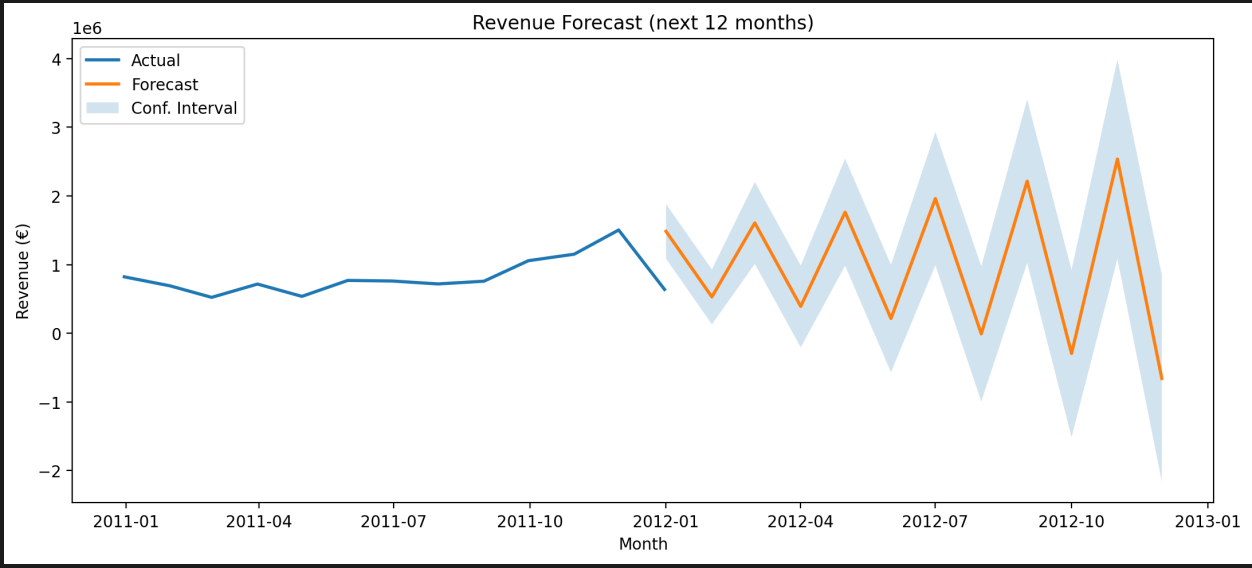
## Cohort Retention

Each cell shows the % of customers who returned after N months, grouped by their first purchase month (cohort).



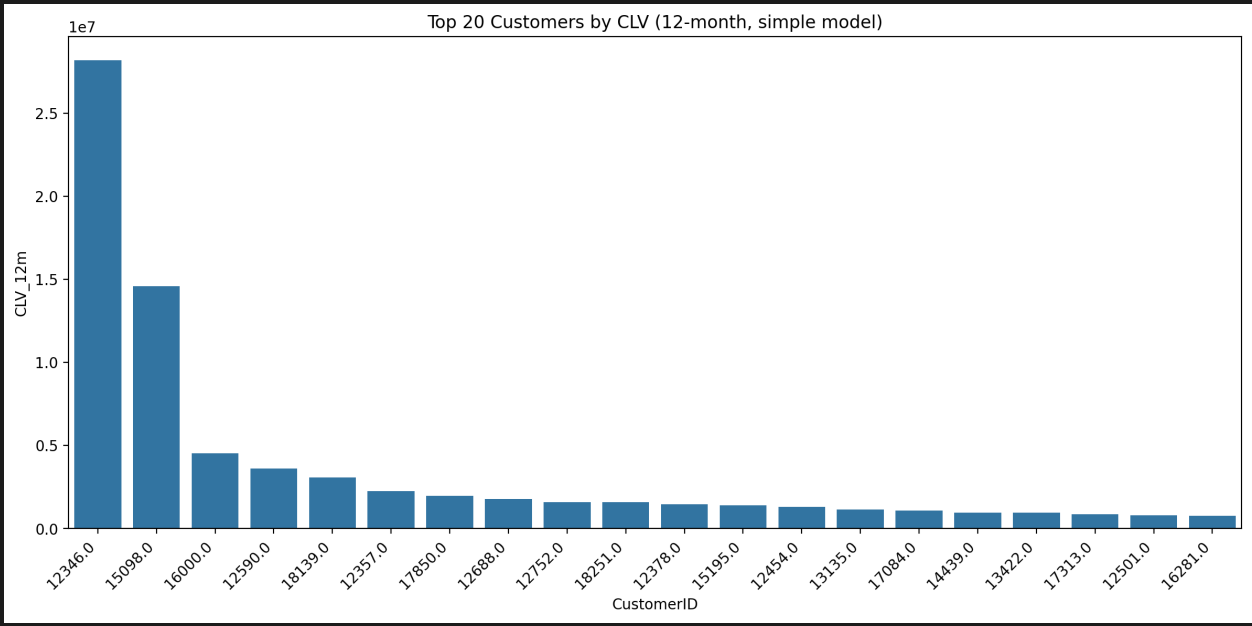
## Revenue Forecast (Next 12 Months)

SARIMAX forecast with 80% confidence band. Use this to plan inventory, marketing budgets, and hiring.



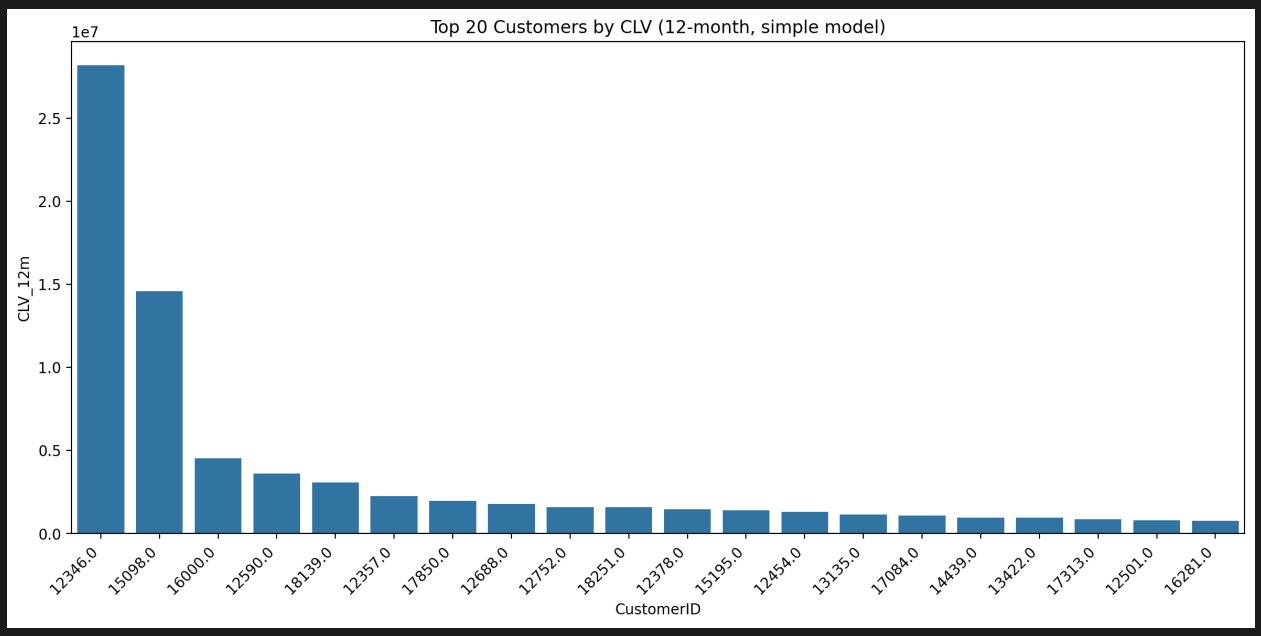
## Customer Lifetime Value (CLV) - 6-Month Outlook

We estimate a simple 6-month CLV as AOV x (avg monthly repeat probability) x 6, added to the historical revenue. Top 20 customers shown.



## Customer Lifetime Value (CLV v1 - 12-Month Model)

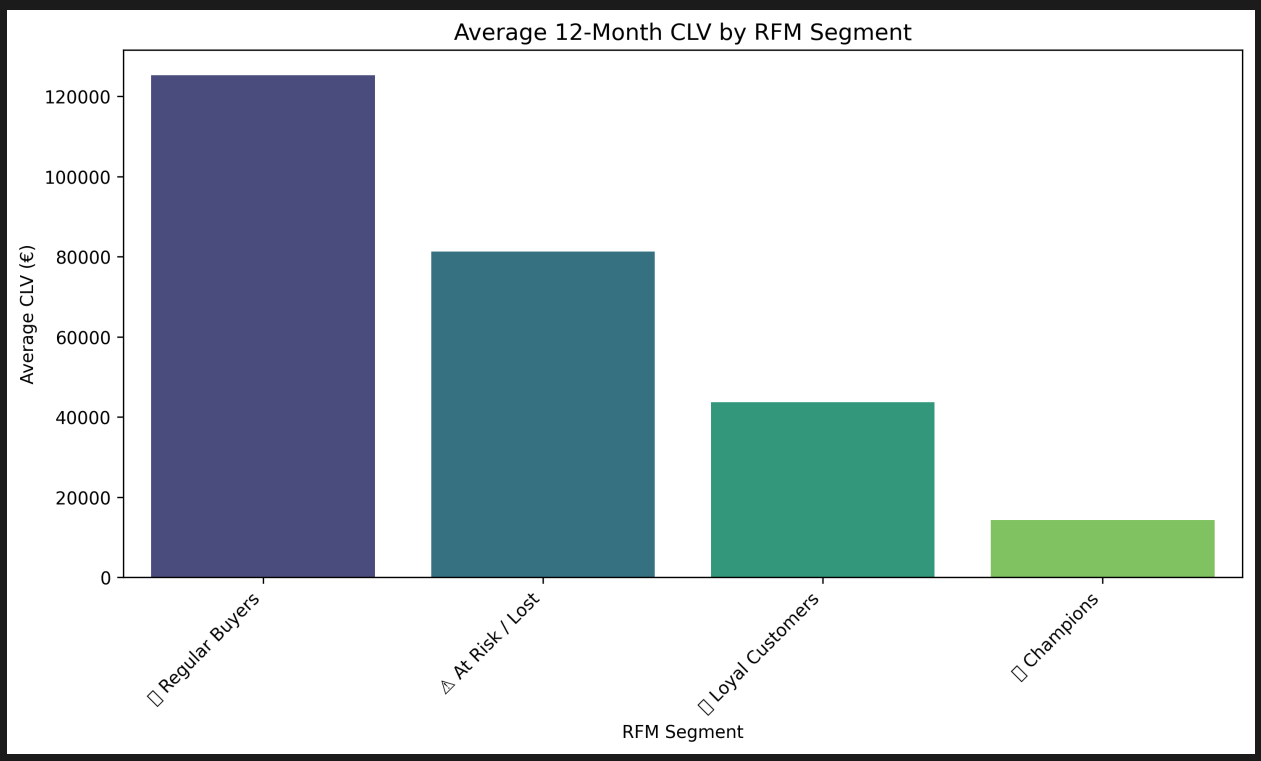
Deterministic CLV v1: AOV x Purchase Frequency per Month x 12 Months. Shows your top-value customers based on average order value and repeat purchase rate.




Last updated (UTC): 2026-10-15 06:53 UTC

## CLV by Customer Segment

Average predicted 12-month CLV for each RFM segment.

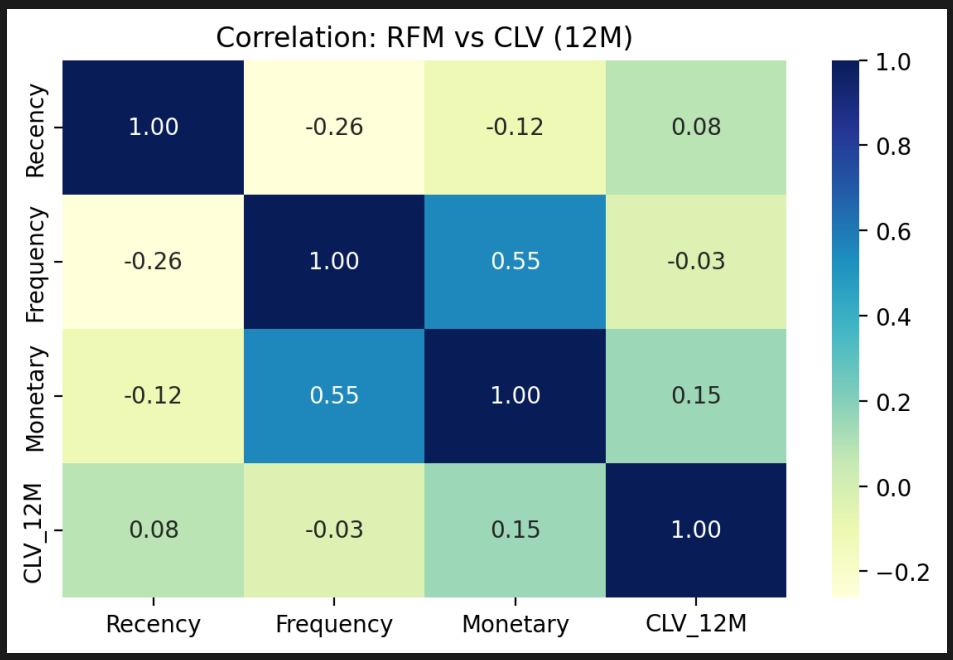


## RFM x CLV - Segment Insights

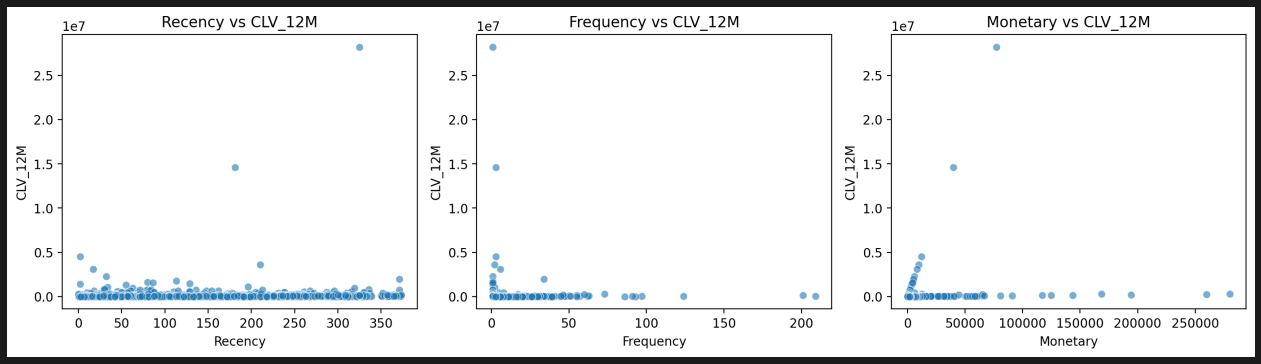
How lifetime value varies by customer segment, and how it correlates with Recency, Frequency and Monetary (RFM).

- ■ **Regular Buyers** - avg CLV â‚¬125,288 (n=1177)
- ■■ **At Risk / Lost** - avg CLV â‚¬81,282 (n=882)
- ■ **Loyal Customers** - avg CLV â‚¬43,711 (n=1026)
- ■ **Champions** - avg CLV â‚¬14,341 (n=1253)

### RFM-CLV Correlation




### CLV vs RFM (example scatter)

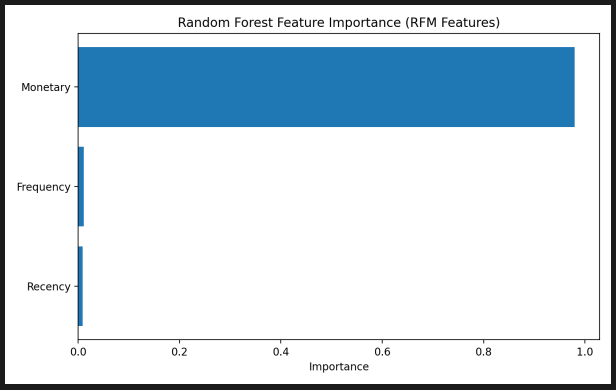


## Predictive CLV Modelling (Machine Learning)

### Model Performance (RFM-Based CLV Prediction)

**Linear Regression**
MAE: 0.00
RMSE: 0.00
$R^2$: 1.00

**Random Forest Regressor**
MAE: 43.66
RMSE: 1052.86
$R^2$: 0.987




### Key Insights

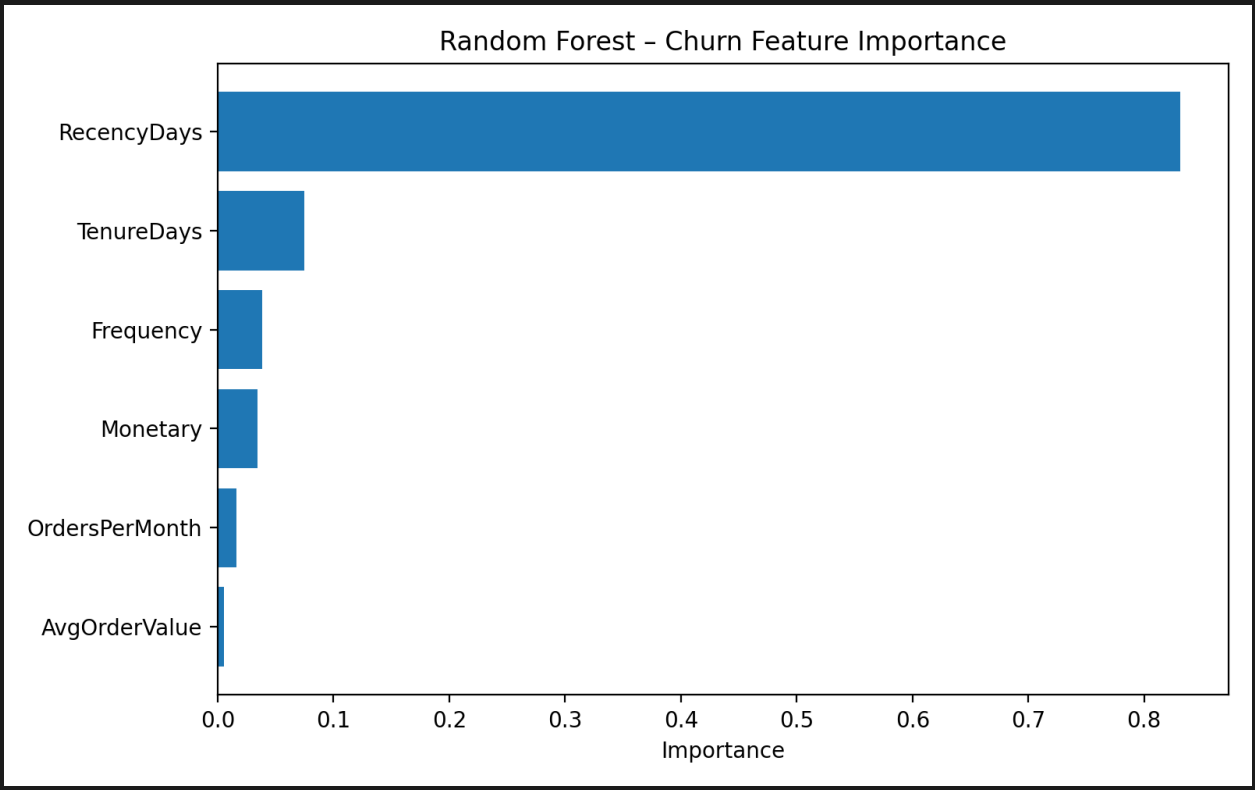
• RFM features (Recency, Frequency, Monetary) are exceptionally strong predictors of Customer Lifetime Value.
• Linear Regression achieved a perfect $R^2 = 1.00$ due to the deterministic relationship between Monetary value and CLV.
• Random Forest ($R^2 = 0.987$) confirms model stability and captures non-linear customer behaviours.
• Recency and Monetary value are the most influential features, reinforcing the validity of the RFM framework.
• Predictive modelling enhances segmentation accuracy and supports proactive customer value strategies.

## Customer Churn Prediction (Phase 10)

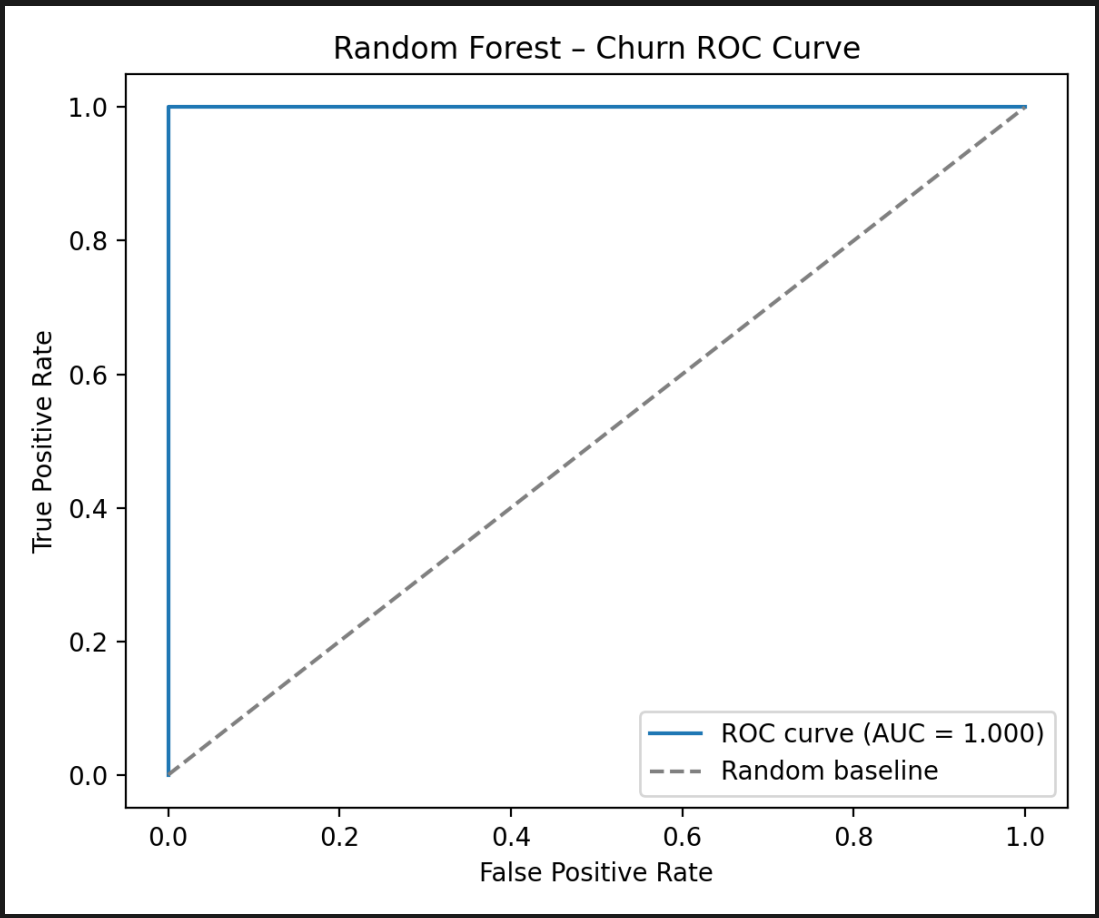
This section introduces a binary classification model predicting whether a customer is likely to become inactive based on purchase behavior. Churn is defined as no purchases in the last 90 days.

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| LogisticRegression | 0.992 | 1.000 | 0.975 | 0.987 | 1.000 |
| RandomForestClassifier | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |

### Random Forest Feature Importance




### Random Forest ROC Curve


Random Forest – Churn ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (AUC = 1.000)
Random baseline